

Random Walk Course

Main Takeaways for Quant Interviews

A compact guide to the ideas, formulas, intuitions, and problem-solving patterns worth retaining.

1. The basic object

A one-dimensional random walk is built from independent increments. If the walk starts at $S_0 = i$, write

$$S_n = i + \sum_{k=1}^n X_k$$

For ± 1 steps, $P(X_k = +1) = p$ and $P(X_k = -1) = q = 1 - p$.

- Fair walk: $p = q = 1/2$, so $E[X_k] = 0$ and $\text{Var}(X_k) = 1$.
- Biased walk: $E[X_k] = p - q = 2p - 1$ and $\text{Var}(X_k) = 4pq$.

$$E[S_n] = i + n(2p - 1), \quad \text{Var}(S_n) = 4npq$$

Intuition: drift determines the mean motion; variance grows linearly in time, so typical displacement from randomness grows like $\sqrt{n}$.

2. First-step analysis: the default tool

When a probability or expected time depends only on the current state, condition on the first step. This creates a recurrence. This is the most reusable random-walk technique.

$$\text{Probability: } u_i = q u_{i-1} + p u_{i+1}$$

$$\text{Expected time: } E_i = 1 + q E_{i-1} + p E_{i+1}$$

- Boundary conditions determine the constants. For absorption at 0 and N, typically $u_0 = 0$, $u^N = 1$ and $E_0 = E^N = 0$.
- Change the boundaries to match the event you actually care about. Example: “return to 4 before hitting 0” is a 0-versus-4 problem, not a 0-versus-5 problem.

Interview habit: before doing algebra, identify the state variable, the absorbing/event boundaries, and what happens after one step.

3. Gambler’s ruin: hitting probabilities

For a fair walk on $\{0, 1, \dots, N\}$, starting at i , the probability of reaching N before 0 is linear in i :

$$P_i(\text{hit } N \text{ before } 0) = i / N$$

With arbitrary boundaries $a < x < b$:

$$P_x(\text{hit } b \text{ before } a) = (x - a) / (b - a)$$

For a biased walk, $p \neq q$:

$$P_i(\text{hit } N \text{ before } 0) = [1 - (q/p)^i] / [1 - (q/p)^N]$$

- The fair formula is “distance from the lower boundary / total interval length.”
- The biased solution is geometric rather than linear because successive differences form a geometric progression.

4. Expected absorption time

For a fair walk absorbed at 0 or N:

$$E_i[T] = i(N - i)$$

- It is largest near the midpoint and zero at the boundaries.
- The quadratic shape is natural: the expected-time recurrence has constant second difference.
- Diffusion intuition: a fair walk moves a typical distance $\sqrt{n}$ in n steps, so traveling a distance L takes order L^2 steps.

For a biased walk, use the same first-step recurrence; the difference equation becomes affine-geometric. A useful algebra trick is to subtract the fixed point of the recurrence before solving.

5. Markov property and decomposing path events

Once the walk reaches a state, the future depends on the present state, not on the detailed path used to get there. This lets you multiply probabilities across sequential stages.

$$P(A \text{ then } B) = P(A) \times P(B \mid \text{state reached after } A)$$

- Example pattern: “hit 4 before 0, then hit 0 before 5.” Solve the two stages separately and multiply.
- For “visit state j exactly once,” first reach j , then after leaving j require absorption before returning.
- For “visit both a and b ,” split according to which one is visited first when useful.

Think in stages. Strong Markov reasoning turns a complicated path condition into several ordinary gambler’s-ruin problems.

6. Return probabilities and number of visits

Inside a finite absorbing interval, repeated visits to an interior state often have a geometric structure.

If r is the probability of returning to state j after leaving it, before absorption, then conditional on reaching j at least once:

$$P(N_j = k \mid \text{reached } j) = r^{k-1}(1 - r), \quad k = 1, 2, \dots$$

$$E[N_j \mid \text{reached } j] = 1/(1 - r)$$

- If the state may never be reached, mix the geometric distribution with a point mass at 0.
- For variances, either use total variance or compute $E[N^2]$ from the conditional geometric distribution and use $\text{Var}(N) = E[N^2] - E[N]^2$.

Important distinction: an unrestricted 1D fair random walk is recurrent, but a walk with finite absorbing boundaries is absorbed almost surely and has finite expected absorption time.

7. Maxima and the tail-sum formula

For a nonnegative integer-valued random variable X :

$$E[X] = \sum_{k=1}^{\infty} P(X \geq k)$$

Best intuition: picture X as the height of a column of unit blocks. Instead of averaging column heights vertically, count the blocks horizontally, level by level. The fraction of columns that reach level k is $P(X \geq k)$.

For a random-walk maximum M , this is especially useful because

$$P(M \geq m) = P(\text{the walk ever hits level } m \text{ before absorption})$$

so an expected maximum becomes a sum of hitting probabilities.

If X has a finite lower bound a (possibly negative), shift the “floor”:

$$E[X] = a + \sum_{k=a+1}^{\infty} P(X \geq k)$$

Mental picture: expected height = baseline + expected number of levels crossed.

8. Conditioning on eventual success changes the walk

If you condition a fair walk on eventually hitting the upper boundary before the lower boundary, the conditioned process is no longer locally fair.

For a fair walk on $\{0, \dots, N\}$, conditioned on eventual success (hit N before 0), from interior state i :

$$P(i \rightarrow i+1 \mid \text{success}) = (i+1)/(2i)$$

$$P(i \rightarrow i-1 \mid \text{success}) = (i-1)/(2i)$$

Intuition: conditioning filters out paths that are incompatible with eventual success, so upward moves become more represented in the surviving sample paths.

Do not keep using $1/2$ - $1/2$ transitions after conditioning on a future event.

9. Martingales

A process M_n is a martingale if, given all information through time n , its expected next value equals its current value:

$$E[M_{n+1} \mid \mathcal{F}_n] = M_n$$

For a fair ± 1 walk, two key martingales are:

$$M_n = S_n$$

$$M_n = S_n^2 - n$$

- For a biased walk, subtract the drift: $S_n - n(2p - 1)$ is a martingale.
- S_n is naturally tied to hitting probabilities.
- $S_n^2 - n$ is naturally tied to expected hitting times.

A martingale is a mathematical model of a fair game: given what you know now, there is no predictable gain in the next step.

10. Optional stopping

If M_n is a martingale and T is a suitable stopping time, then under the required regularity conditions:

$$E[M_T] = E[M_0]$$

A stopping time can depend on the past and present, but cannot look into the future. “Stop the first time $S_n = 5$ ” is valid; “stop at the last time $S_n = 0$ ” is not.

The proof idea worth remembering:

1. Freeze the process at T : $M_n \wedge T$.
2. The stopped process is still a martingale.
3. For each fixed n , $E[M_n \wedge T] = E[M_0]$.

4. Under suitable conditions, let $n \rightarrow \infty$ to obtain $E[M_T] = E[M_0]$.

In finite gambler's ruin, the stopped position is bounded, so optional stopping is safe.

Core intuition: a fair game does not acquire positive expected value merely because you choose when to stop—provided the stopping rule satisfies the theorem's conditions.

11. The “high win rate, rare disaster” intuition

Suppose a fair walk starts at 0 and you stop at +100 or −1000. The probability of hitting +100 first is 10/11, while the probability of hitting −1000 first is 1/11.

$$E[S_T] = (10/11)(100) + (1/11)(-1000) = 0$$

So a strategy can win about 91% of the time and still have zero expected profit. The rare loss is large enough to offset the frequent modest wins.

This is the right intuition for why stopping does not create value. It can reshape the distribution of outcomes without changing the expectation.

12. When optional stopping fails / infinite-expectation traps

Do not apply optional stopping blindly. A classic trap is an unrestricted fair walk starting at 0 and stopping the first time it hits +1. It hits +1 almost surely, so $S_T = 1$, yet $E[S_0] = 0$.

There is no contradiction: before hitting +1, the walk has unlimited downside and can wander arbitrarily far negative. The boundedness / convergence conditions used in finite gambler's ruin fail.

- Almost sure finiteness of T alone is not enough to justify optional stopping.
- A fair walk started at 1 hits 0 almost surely, but its expected hitting time is infinite.
- Habit: before computing an expectation, ask whether the stopping event occurs almost surely and whether the expectation is finite.

13. Useful interview patterns

Problem signal	First thought
Hitting probability	Use gambler's ruin or a first-step recurrence.
Expected absorption time	Use $E_i = 1 + qE_{i-1} + pE_{i+1}$, or $S_n^2 - n$ + optional stopping in the fair case.
Maximum / range	Turn the expectation into tail probabilities, then convert $P(M \geq m)$ into a hitting probability.
Exactly once / at least twice	Reach the state first, then analyze return-before-absorption probabilities.
Number of visits	Look for a geometric return structure; mix with probability of ever reaching the state.
Sequential path event	Break into stages and use the Markov property.
Conditioned path	Recompute transition probabilities under the conditioning.

Problem signal	First thought
Suspicious stopping problem	Check for unbounded downside or infinite expected stopping time before invoking optional stopping.

14. Formula sheet: worth remembering

Item	Formula
Fair hitting probability	$P_i(\text{hit } N \text{ before } 0) = i/N$
Fair arbitrary boundaries	$P_x(\text{hit } b \text{ before } a) = (x-a)/(b-a)$
Biased hitting probability	$[1-(q/p)^i]/[1-(q/p)^N], \quad p \neq q$
Fair expected absorption time	$E_i[T] = i(N-i)$
Tail sum	$E[X] = \sum_{k \geq 1} P(X \geq k), \text{ for nonnegative integer } X$
Finite geometric sum	$\sum_{k=0}^{m-1} r^k = (1-r^m)/(1-r)$
Fair-walk martingales	S_n and $S_n^2 - n$
Biased drift-corrected martingale	$S_n - n(2p-1)$
Geometric mean / variance	$E[G]=1/p, \quad \text{Var}(G)=(1-p)/p^2$

15. What to derive rather than memorize

- The first-step recurrences themselves: derive them from conditioning on the next step.
- Return probabilities: solve the small gambler's-ruin problem that matches the relevant return event.
- Visit-count distributions: identify the probability of another return, then recognize a geometric structure.
- Most special-case maxima: use tail sum + hitting probability rather than memorizing a bespoke formula.
- Optional-stopping applications: remember the martingales and the principle; do not memorize a long theorem statement.

Bottom line: random-walk interview problems are mostly about choosing the right representation — first-step recursion, gambler's ruin, Markov decomposition, tail sums, or a martingale — rather than doing difficult algebra.

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